

At Larger Radius the Occupancy-Stable First-Law Probe Dies

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Abstract

Paper 29 left C4 (CHM parabola versus linear $R - r$) as a tie on a ball of radius 4. A larger ball is the next place those envelopes can split. It is not a step toward the Planck length.

$N = 14$, $R = 5$, thirty locked surface hops. At $\varepsilon = 0.01$, 27 of 30 flip n_{occ} . At $\varepsilon = 0.002$, 25 of 30 still flip. The five survivors are not a census. C4 is not scored.

The last valid uniqueness test remains Paper 29: 294 hops, zero flips, a 0.002 residual tie. This note records an instrument halt, not a nearer ultraviolet.

1 Why it dies

The unperturbed sea has $n_{\text{occ}} = 1370$ on $14^3 = 2744$ sites. Half filling is 1372. A surface hop of size 0.002 moves a near-zero mode across the Fermi level. That is not linear Clausius. It is a level crossing.

Admission. *I went to larger R to split the parabola from a linear weight. The probe that was stable at $R = 4$ is not stable at $R = 5$. I did not call that the Planck scale, and I did not score five leftover hops as a theorem.*